EPAData Project Phase 2 Grading Rubric

This rubric evaluates steady project development, the working Phase 2 application, and the final written report. The assessment emphasizes evidence of implementation and understanding across data analysis, EPA TRACI impact assessment, interactive visualization, prediction, and sensitivity analysis.

Assessment Summary

Assessment Area	Points	Weight
Project progress	20	20%
Demonstration and presentation	60	60%
Project report	20	20%
Total	100	100%

1 Project Progress

20 points. Progress is evaluated from weekly meetings and the team's GitHub development record.

1.1 Weekly Meetings

Six weekly meetings are worth 2 points each, for a total of 12 points.

Performance	Points per Week	Evidence
Excellent	2.0	Team attends prepared, demonstrates meaningful progress, explains decisions and problems, and presents a realistic plan for the following week.
Proficient	1.5	Team reports progress and next steps, but the evidence or explanation lacks some detail.
Developing	1.0	Limited progress is shown, preparation is weak, or important team members are absent.
Insufficient	0–0.5	Meeting is missed, or the team cannot demonstrate meaningful progress.

Expected Six Week Progress

Week	Milestone
1	Project definition and analytical data layer
2	Descriptive and EPA TRACI impact analysis
3	Interactive visualization
4	Prediction models
5	Sensitivity and scenario analysis
6	Integration, testing, and final delivery

1.2 GitHub Development Records

Criterion	Points	Full Credit Standard
Commit history	3	Regular, meaningful commits are distributed across the six weeks, and commit messages clearly describe changes.
Team participation	2	GitHub records show reasonable contributions from all team members.
Branch and merge practices	1	Features are developed safely through branches or pull requests without damaging working code.
Issues and documentation	1	Tasks, bugs, milestones, or design decisions are recorded in issues, pull requests, or project documentation.
Repository quality	1	No credentials or unnecessarily large raw datasets are committed; README and environment instructions are current.

Note: A large number of commits made only immediately before the deadline does not demonstrate steady project progress.

2 Demonstration and Presentation

60 points. Students must demonstrate the working application and explain the design, methods, results, and limitations.

Criterion	Points	Full Credit Evidence
Analytical data layer	7	● Uses multiple years of CAMPD data and the selected TRACI factors ● Handles missing or invalid values appropriately ● Supports filtering, aggregation, and derived indicators ● Preserves relevant units and data-source information
Descriptive and comparative analysis	7	● Provides appropriate descriptive statistics and grouped comparisons ● Calculates emission intensity, heat rate, or other selected indicators correctly ● Handles missing values and division by zero correctly ● Interprets results rather than merely displaying numbers
EPA TRACI impact assessment	8	● Matches CAMPD pollutants with verified TRACI air-emission factors ● Converts short tons to kilograms and applies the impact formula correctly ● Keeps impact categories and units separate and records the TRACI version ● Explains that results are potential-impact indicators, not direct health outcomes
Interactive visualization	10	● Includes at least two meaningful visualization types ● Uses clear titles, axes, units, legends, labels, and tooltips ● Supports useful filtering or interaction and image or data export ● Students explain what each visualization shows and why it is appropriate
Prediction module	9	● Defines a clear target and prevents target leakage ● Compares a simple baseline with at least two candidate models ● Uses an appropriate training and testing strategy and evaluation metrics ● Explains performance, errors, and limitations without claiming causation
Sensitivity and scenario analysis	8	● Provides one-variable-at-a-time analysis with other inputs held constant ● Calculates absolute and percentage changes from a clearly identified baseline ● Displays a sensitivity curve and compares at least two scenarios ● Warns when inputs are outside the training range and explains model-response limits
Application integration and usability	5	● Integrates Phase 2 functions into the existing Flask application ● Provides understandable navigation, input validation, and error handling ● Produces reproducible results and runs from the submitted instructions
Presentation quality and participation	6	● States the analytical question and organizes the presentation logically ● Demonstrates the application live and explains technical decisions clearly ● All team members participate, discuss limitations, and answer questions

3 Project Report

20 points. The report must accurately document the submitted application, methods, evidence, and limitations.

Criterion	Points	Full Credit Evidence
Objectives and data sources	3	Defines the analytical and prediction questions; Describes CAMPD and TRACI data, reporting years, and selected variables; Explains how the two datasets are connected.
Methods	5	Explains data preparation, missing-value handling, and derived indicators; Documents TRACI matching, unit conversion, and impact calculations; Explains visualization, prediction, evaluation, and sensitivity procedures.
Results and interpretation	5	Presents important descriptive, comparative, and TRACI impact results; Includes clearly labeled tables or figures; Compares prediction models with the baseline and interprets sensitivity results; Connects findings to the original project questions.
Evaluation limitations and validation	4	Evaluates model performance appropriately; Discusses data quality, uncertainty, model limits, and TRACI limitations; Distinguishes association, prediction, and causation; Describes relevant software tests or calculation validation.
Organization reproducibility and writing	3	Uses logical organization, professional writing, correct terminology, and units; Cites EPA CAMPD, EPA TRACI, software, and other sources; Provides enough information to reproduce the analysis; Remains consistent with the submitted application and repository.

4 Performance Levels

Apply these performance levels within each demonstration or report criterion.

Level	Approximate Credit	General Description
Excellent	90–100%	Complete, correct, well integrated, clearly explained, and supported by convincing evidence.
Proficient	75–89%	Mostly complete and correct, with minor omissions or explanation problems.
Developing	50–74%	Partially implemented or explained, with important omissions, errors, or integration problems.
Insufficient	0–49%	Missing, largely nonfunctional, seriously incorrect, or not understood by the team.

5 Important Grading Conditions

- A demonstration consisting only of slides, screenshots, or prerecorded output does not receive full demonstration credit.
- Features must operate using the submitted project code and documented environment.
- A student who cannot explain a claimed contribution may receive an individual score adjustment.
- API keys, passwords, and other credentials must not be committed to GitHub.
- Using AI tools or open-source libraries does not replace the requirement to understand and explain the submitted implementation.